Customer Shopping Behavior Analysis

Client Project Report

A concise business-focused report covering customer transaction data preparation, segmentation, database integration, and SQL analysis.

Project Scope	Customer shopping behavior
Records Analyzed	3,900
Data Fields	18 original fields
Analysis Stack	Python / Pandas / NumPy / PostgreSQL / SQL

Prepared based on the supplied analysis workflow and SQL deliverables.

1. Executive Summary

The project analyzes a customer shopping dataset containing 3,900 purchase records and 18 fields. The work covers data inspection, quality checks, customer segmentation, promotional classification, database integration, and structured SQL analysis.

The analysis is designed to provide a reusable foundation for understanding customer purchasing patterns, customer segments, product/category performance, payment behavior, promotion usage, and purchase frequency.

3,900	18	$59.76	3.75 / 5
Purchase records	Original fields	Average purchase	Average rating

Key Data Indicators

Indicator	Observed value
Customer age range	18–70 years
Average customer age	44.07 years
Average previous purchases	25.35
Median purchase amount	$60
Purchase amount range	$20–$100
Missing review ratings	37 records

2. Data Review & Quality Assessment

The source data was loaded and reviewed before analysis. The dataset contains customer demographics, product information, transaction values, subscription information, promotional indicators, shipping, payment method, review rating, and purchase frequency.

Data Quality Check

The dataset contains 3,900 complete records for most fields. The only missing values identified in the provided analysis are 37 missing entries in Review Rating. All other 17 original fields contain complete records.

Data quality area	Result
Total records	3,900
Total original fields	18
Fields with missing values	Review Rating only
Missing Review Rating values	37
Other missing values	None identified

Descriptive Statistics

The initial statistical review shows an average purchase amount of $59.76, with a median of $60. Purchase values range from $20 to $100. The average customer age is 44.07 years, while the average number of previous purchases is 25.35.

3. Customer Segmentation

Two analytical segmentation fields were created to make the customer data easier to interpret and use in downstream reporting.

Age Group

Segment	Age range
Teenager	0–17
Adult	18–44
Middle-aged	45–59
Senior	60+

Loyalty Segment

Customers were grouped according to their Previous Purchases field, creating a practical loyalty framework.

Segment	Previous purchases
Low Loyalty	0–10
Regular	11–25
Loyal	26–40
Highly Loyal	41+

These segments provide a consistent basis for comparing purchasing behavior across different customer groups.

4. Promotion & Purchasing Analysis

A unified **Promotion Used** indicator was created from Discount Applied and Promo Code Used. A customer is classified as 'Used' when either promotional indicator is marked Yes.

SQL Analysis Prepared

The accompanying SQL deliverable provides structured queries for business analysis across the processed customer table.

Analysis area	Business view
Customer KPIs	Total customers, total purchase amount, average purchase amount
Customer experience	Average review rating
Customer history	Average previous purchases
Subscription	Customer count by subscription status
Category performance	Purchase amount by category
Product performance	Purchase amount by item, ranked highest to lowest
Seasonality	Purchase amount by season
Payment behavior	Payment method usage
Demographics	Purchase amount by age group and gender
Promotion	Customers by promotion usage
Loyalty	Customers by loyalty segment
Purchase frequency	Purchase amount by purchase-frequency group

5. Database Integration & Deliverables

The transformed customer dataset was loaded into a PostgreSQL database using Python and SQLAlchemy. This creates a structured database layer for repeatable SQL reporting and future dashboard development.

Component	Delivered
Python analysis	Data loading, inspection and transformation workflow
Customer segmentation	Age Group and Loyalty Segment
Promotion classification	Promotion Used indicator
Database	PostgreSQL customer_behavior database
Data table	Customer
SQL analysis	Business-focused KPI and segmentation queries

Overall Workflow

Source Dataset → **Data Validation** → **Feature Creation** → **PostgreSQL** → **SQL Business Analysis**

Client Value

- Creates a clean, structured base for customer analytics and reporting.
- Provides standardized customer segments for targeted analysis.
- Converts raw promotional fields into a business-friendly indicator.
- Makes the dataset available in PostgreSQL for repeatable SQL analysis.
- Establishes a foundation for future dashboards and deeper customer insights.

6. Conclusion

The completed work establishes a practical customer analytics pipeline from raw transactional data through structured business analysis. The dataset has been inspected, quality-checked, transformed into useful customer segments, and prepared for SQL-based reporting.

The current deliverables provide the foundation for further management reporting, including interactive visual dashboards, category and product performance views, customer segment comparisons, promotion analysis, and purchase-frequency reporting.

Deliverables Included

- Processed Python/Jupyter analysis workflow
- PostgreSQL database integration
- SQL business analysis queries
- Customer segmentation framework
- Client-ready project report

Note: This report presents only findings and capabilities supported by the supplied notebook and SQL deliverables. SQL query definitions were provided, but their result tables were not included in the supplied SQL file; therefore, no SQL output values have been invented in this report.